

Homework — 1.1.2 Types of Processor

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 1.1.2 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Complete the comparison table of **RISC** and **CISC** processors by filling in the missing cells. **[3]** (AO1)

Feature	RISC	CISC
Size of instruction set	Small set of simple instructions	(a)
Instruction length	(b)	Variable length
Clock cycles per instruction	(c)	Often several cycles
Where the complexity lies	(d)	In the hardware/circuitry
Program length / RAM needed	(e)	Shorter programs, less RAM needed
Typical power consumption	Lower	(f)

2. Match each term to its definition by writing the correct **letter** in the Match column. Two definitions are not needed. **[3]** (AO1)

Term	Match
Multicore processor	
Parallel processing	
GPU	
Co-processor	
Pipelining	
RISC	

Letter	Definition
A	An extra processor that relieves the CPU of specific tasks, e.g. floating-point arithmetic or graphics
B	Carrying out several instructions/tasks at the same time
C	A processor with a very large number of simpler cores, designed to apply the same operation to many items of data at once
D	A CPU containing two or more processing units, each able to fetch, decode and execute its own instructions
E	A processor design with a small set of simple, fixed-length instructions that typically execute in one clock cycle
F	Overlapping the fetch, decode and execute stages of consecutive instructions
G	An architecture in which one shared memory holds both instructions and data
H	Small, very fast memory located on or near the CPU

3. Each statement in the table below contains **one** factual error about GPUs. Identify the error in each and rewrite the statement correctly on the lines below. **[3]** (AO1)

#	Statement (each contains one error)
1	A GPU contains a small number of highly complex cores — far fewer than a typical CPU.
2	A GPU processes graphics efficiently because each core carries out a different instruction on the same single item of data.
3	GPUs cannot be used for any task other than producing graphics.

.....

.....

.....

.....

4. A smartphone manufacturer uses a **RISC (ARM)** processor rather than a **CISC (x86)** processor. Give **two** reasons, linked to a smartphone, why RISC is the better choice. **[2]** (AO2)

.....

.....

.....

5. Tick **True** or **False** for each statement. If a statement is false, write a correction in the final column. **[3]** (AO1/AO2)

Statement	True / False	Correction if false
Any program automatically runs faster on a processor with more cores.		
A co-processor system uses an additional processor to take over specific tasks, such as floating-point calculations, from the CPU.		
In parallel processing, the computer still completes every instruction one after another, in strict sequence.		

Part B — Retrieval: keep it fresh (~6 marks)

6. Explain why the **Von Neumann** architecture can suffer from a performance **bottleneck**, and state how the **Harvard** architecture avoids it. [3] (1.1.1)

7. State **two** factors (other than number of cores) that affect the performance of a CPU, and for each give its effect. [2] (1.1.1)

8. Name the register that holds the **current instruction** while it is being decoded and executed. [1] (1.1.1)
